

PySimPace v2.0: An Easy-to-Use Simulation Tool with Machine Learning Pipelines for Realistic MRI Motion Artifact Generation

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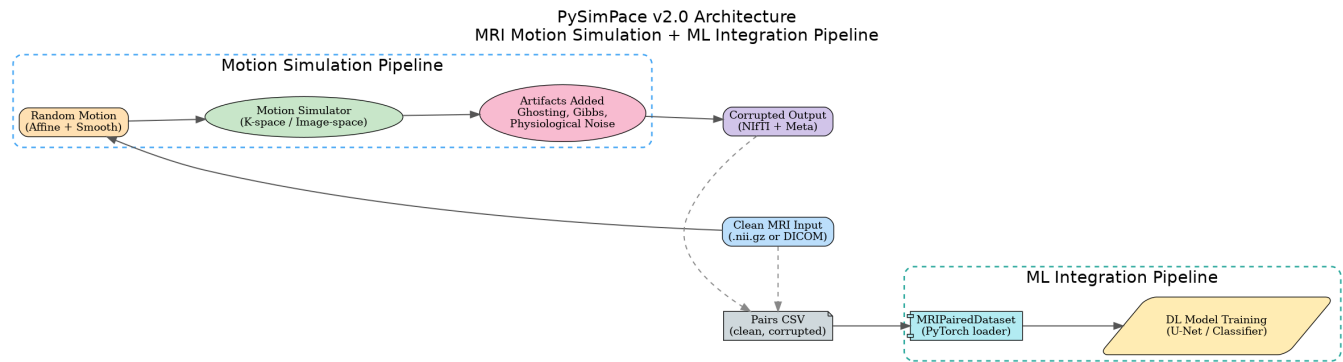


Figure 1: Architecture of py-simpac v2.0: clean MRI data is processed through motion simulation and artifact generation pipelines to produce paired data for ML training.

Abstract

Motion artifacts in structural and functional magnetic resonance imaging (MRI) pose a significant challenge for both clinical use and machine learning (ML)-based image analysis. Existing ML approaches for artifact correction require paired clean and corrupted datasets, which are difficult to acquire. We present **py-simpac**, an open-source, pip-installable MRI motion artifact simulation toolkit with native ML integration. py-simpac supports structural MRI and functional MRI (fMRI) simulation, offering configurable k-space and image-space motion, ghosting, Gibbs ringing, and physiological noise. It provides an end-to-end pipeline with a ready-to-use PyTorch Dataset interface for ML training. We describe the design

of py-simpac v2.0, compare it with existing tools, and demonstrate its utility for robust artifact correction model development.

CCS Concepts

• **Applied computing** → **Imaging**; • **Computing methodologies** → **Machine learning**.

Keywords

MRI Simulation, Motion Artifacts, fMRI, Machine Learning Pipelines, PyTorch, Open Source Software, k-space Motion, Image-space Motion, Artifact Generation

ACM Reference Format:

Snehil Kumar, Neil Vaughan, Zeyu Fu, and Heather Wilson. 2025. PySimPace v2.0: An Easy-to-Use Simulation Tool with Machine Learning Pipelines for Realistic MRI Motion Artifact Generation. In *Proceedings of the 33rd ACM International Conference on Multimedia (MM '25)*, October 27–31, 2025, Dublin, Ireland. ACM, New York, NY, USA, 4 pages. <https://doi.org/10.1145/3746027.3756875>

1 Introduction

Motion artifacts in structural and functional MRI degrade image quality and hinder analysis. Even submillimeter motion can cause blurring, ghosting, and spurious correlations in fMRI [12].

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MM '25, Dublin, Ireland

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ACM ISBN 979-8-4007-2035-2/2025/10

<https://doi.org/10.1145/3746027.3756875>

Recent machine learning (ML) methods show promise for MRI artifact correction, but training robust models requires large paired datasets of clean and motion-corrupted images, which are hard to obtain due to ethical, practical, and cost constraints [3].

Synthetic data generation is therefore becoming a key strategy for developing ML models for motion correction. Several existing tools provide artifact simulation capabilities — including SimPACE [3], TorchIO [11], and view2Dmotion [8], but these solutions have limitations, such as limited artifact diversity, partial fMRI support, or lack of integration with ML pipelines.

We introduce **py-simpac**, an open-source, pip-installable MRI motion artifact simulation toolkit that enables realistic artifact generation with seamless machine learning integration. **py-simpac** supports both structural MRI (3D) and fMRI (4D), offering configurable k-space and image-space motion, physiological noise, ghosting, and Gibbs ringing artifacts. The toolkit provides an end-to-end pipeline, including a PyTorch-compatible MRIPairedDataset for model training.

py-simpac is designed to be fast, flexible, and extensible, lowering the barrier for MRI researchers and ML practitioners to generate high-quality training data for motion artifact correction.

The key features of **py-simpac** v2.0 include:

- **Realistic motion simulation:** Supports both image-space (slice-wise) motion and blended k-space motion to model smooth, continuous head motion.
- **Artifact generation:** Includes configurable ghosting, Gibbs ringing, and physiological noise models.
- **Full ML integration:** Provides a `generate_training_pairs` utility and `MRIPairedDataset` class for seamless integration with PyTorch-based training pipelines.
- **Support for both structural MRI and fMRI:** Enables intra-volume and inter-volume motion simulation for 3D and 4D data.
- **Ease of use:** Simple API and CLI (planned) to enable fast adoption by researchers and students, with extensive documentation and tutorials.
- **Extensibility:** Modular architecture to facilitate addition of new artifact types and simulation strategies.

In this paper, we present the design and features of **py-simpac** v2.0, compare it with existing solutions, and demonstrate its utility in the development of robust deep learning models for MRI artifact correction.

2 Related Work and Motivation

Motion artifacts degrade MRI quality, impacting both clinical diagnosis and downstream analysis. While retrospective correction methods exist, they often fail under large intra-scan motion [12]. DL-based models offer promising alternatives [1, 7, 14], but require paired data for supervised training [3].

Simulation-based methods are gaining traction:

- **SimPACE** [3]: Generates intra-volume motion using special pulse sequences; lacks ML compatibility.
- **TorchIO** [11]: Offers flexible augmentation but no built-in paired dataset generation; limited fMRI support.
- **view2Dmotion** [8]: Demonstrates slice-level motion simulation; not suited for full-volume DL pipelines.

Other works focus on motion correction via GANs [2, 9, 16], clinical denoising [13], and systematic reviews [4, 5]. However, most datasets [6, 10, 15] lack controlled motion labels, making simulation essential.

py-simpac addresses these gaps:

- Dual support for structural and fMRI motion simulation.
- Generation of paired datasets with PyTorch-ready loaders.
- Open-source, extensible, and pip-installable for broad adoption.

By unifying artifact simulation with ML training workflows, **py-simpac** empowers robust model development and reproducible research in MRI motion correction.

3 Software Design and Implementation

py-simpac v2.0.1 is implemented as a modular, extensible Python package. The core architecture is illustrated in Figure 1.

The pipeline consists of the following components:

- **Motion simulation:** Supports both image-space (slice-wise) and blended k-space motion simulation. Blended k-space motion is a key contribution that generates realistic smooth motion effects across volumes.
- **Artifact generation:** Includes ghosting, Gibbs ringing, and physiological noise models. Each can be selectively enabled and parametrized.
- **ML integration:** Generates paired clean / corrupted datasets ready for training. Provides a built-in `MRIPairedDataset` PyTorch-compatible dataset class.
- **Extensibility:** The architecture is modular and supports new artifact models to be added with minimal effort.

The tool supports both 3D (structural MRI) and 4D (fMRI) pipelines. A command-line interface (CLI) and GUI are under development.

Installation: **py-simpac** is fully open-source and available on PyPI. It can be installed with:

```
pip install py-simpac
```

The source code and examples are available on GitHub:

```
git clone https://github.com/snehil03july/py-SimPace.git
cd py-SimPace
pip install -e .
```

Dependencies:

- `numpy`, `scipy`, `matplotlib`, `nibabel`, `torch`, `pandas`, `tqdm`, `pyyaml`.

4 Evaluation and Comparison with Existing Tools

We compare **py-simpac** with existing open-source MRI motion simulation tools in Table 1. The key advantages of **py-simpac** are its support for both structural and fMRI pipelines, realistic artifact models, and direct ML pipeline integration.

5 Usage Example and Case Study

We demonstrate usage of **py-simpac** on structural MRI data.

First, a clean MRI volume is processed using `simulate_structural_motion()` with selected artifact parameters.

Table 1: Feature comparison of py-simpac with existing open-source MRI motion artifact simulation tools. ✓ = Supported, ✗ = Not supported.

Feature	py-simpac v2.0.1 (this work)	SimPACE [3]	TorchIO [11]	view2Dmotion [8]
Structural MRI support	✓	✗	✓	✓
fMRI support	✓	✓	Partial	✗
Rigid motion simulation	✓	✓(slice)	✓	✓(2D)
Ghosting artifact	✓	✗	✓	✓
Gibbs ringing artifact	✓	✗	✗	✗
Physiological noise (fMRI)	✓	✗	✗	✗
ML pipeline integration	✓	✗	✗	✗
Paired dataset generation	✓	✗	✗	✗
PyTorch-ready dataset class	✓	✗	✗	✗
CLI support	✓	✗	✗	✗

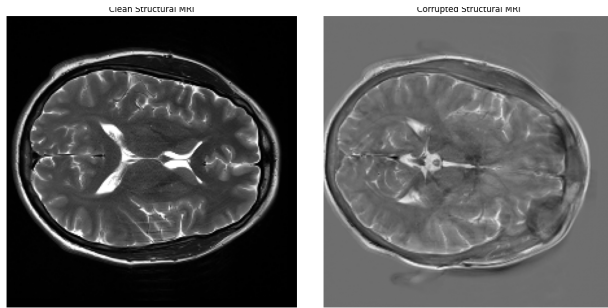
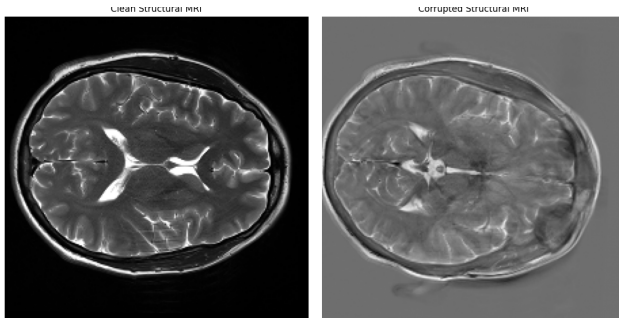
**Figure 2: Example of Gibbs ringing artifact simulated in structural MRI using py-simpac.****Figure 3: Example of Ghosting artifact simulated in structural MRI using py-simpac.**

Figure 2 shows an example of a **Gibbs ringing artifact**. Figure 3 shows an example of a **Ghosting artifact**.

Generated paired datasets can be easily loaded using:

Listing 1: Example usage of MRIPairedDataset from py-simpac.

```
from pysimpac.ml import MRIPairedDataset
dataset = MRIPairedDataset("examples/training_pairs/pairs.csv")
```

Reproducibility Note. To ensure reproducibility, we provide a pre-configured Dockerfile that builds a fully isolated environment with all dependencies for **py-simpac**. Researchers can quickly build and run the toolkit by executing the Docker commands provided in the README, enabling seamless integration into pipelines and eliminating environment inconsistencies. The Docker image includes example scripts and a working demo for rapid testing and validation.

Code and Reproducibility

The full source code, example scripts, documentation, and Dockerfile for **py-simpac v2.0** are publicly available on GitHub at: <https://github.com/snehl03july/py-SimPace>. While the code is publicly accessible via GitHub, we also provide a direct ZIP archive for convenience: <https://drive.google.com/file/d/1Aezy8yNc5tHmBmxuTepsPVAJmaSmZcLy/view?usp=sharing>

6 Conclusion and Future Work

We presented **py-simpac**, a modular open-source Python library for realistic MRI motion artifact simulation with built-in machine learning integration.

Key features include support for both structural and functional MRI, generation of diverse artifacts (ghosting, Gibbs ringing, physiological noise), and seamless creation of PyTorch-ready paired datasets. Its scalable and extensible design allows fast and customizable data generation.

Planned extensions include:

- New artifacts: spike noise, intensity drift, susceptibility distortions, and partial Fourier artifacts.
- A user-friendly GUI (React/Streamlit frontend + Python backend) for interactive use.
- CLI tools for large-scale automated dataset generation.

py-simpac is now available on PyPI and GitHub, and we invite the community to contribute toward its continued development and adoption.

Availability

- GitHub: <https://github.com/snehl03july/py-SimPace>

- PyPI: <https://pypi.org/project/py-simpace/>
- Install via: `pip install py-simpace`

Acknowledgments

This research was supported by the Engineering and Physical Sciences Research Council (EPSRC) through a doctoral training grant. We thank the University of Exeter's Clinical and Biomedical Sciences (CBS) and Computer Science departments for their support. For the purpose of open access, the author has applied a Creative Commons Attribution (CC BY) licence to any Author Accepted Manuscript version arising. The research was carried out at the National Institute for Health and Care Research (NIHR) Exeter Biomedical Research Centre (BRC).

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